

Fire risk prediction in the Paraguayan Chaco region

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ABSTRACT

While wildfires naturally occur in various landscapes, their impacts on fauna, flora, ecosystems, and communities are intensifying globally due to climate change and human-driven land use changes (Vidal-Riveros et al. 2024, Bera et al. 2025). These fires affect biodiversity, soil ecology, carbon storage, and habitat quality, while also worsening other disturbances to key ecosystem services, ultimately degrading the living conditions of both human settlements and wildlife (Vidal-Riveros et al. 2024, Busari et al. 2025). In Paraguay, the Chaco region experiences the largest concentration of wildfires, accounting for over 50% of the country's total fires (Coronel et al. 2021). Among wildfire management strategies, accurate early prediction helps reduce wildfire damage and lowers logistical costs associated with fire suppression (Honary et al. 2025). The goal of this project is to develop an empirical model to predict wildfire likelihood in the Paraguayan Chaco, aiding response teams in preparedness for major fire events. Bera et al. 2025 applied a random forest method to create a 5-day wildfire susceptibility map using climatic data from the previous six days, combined with vegetation and topographic information. This project will adapt that method to the Paraguayan Chaco by incorporating a land cover variable and generating proximity rasters based on country-specific layers. Wildfire occurrences from 2015 to 2024 will serve as validation data, alongside various raster datasets used as explanatory variables.

OBJECTIVES

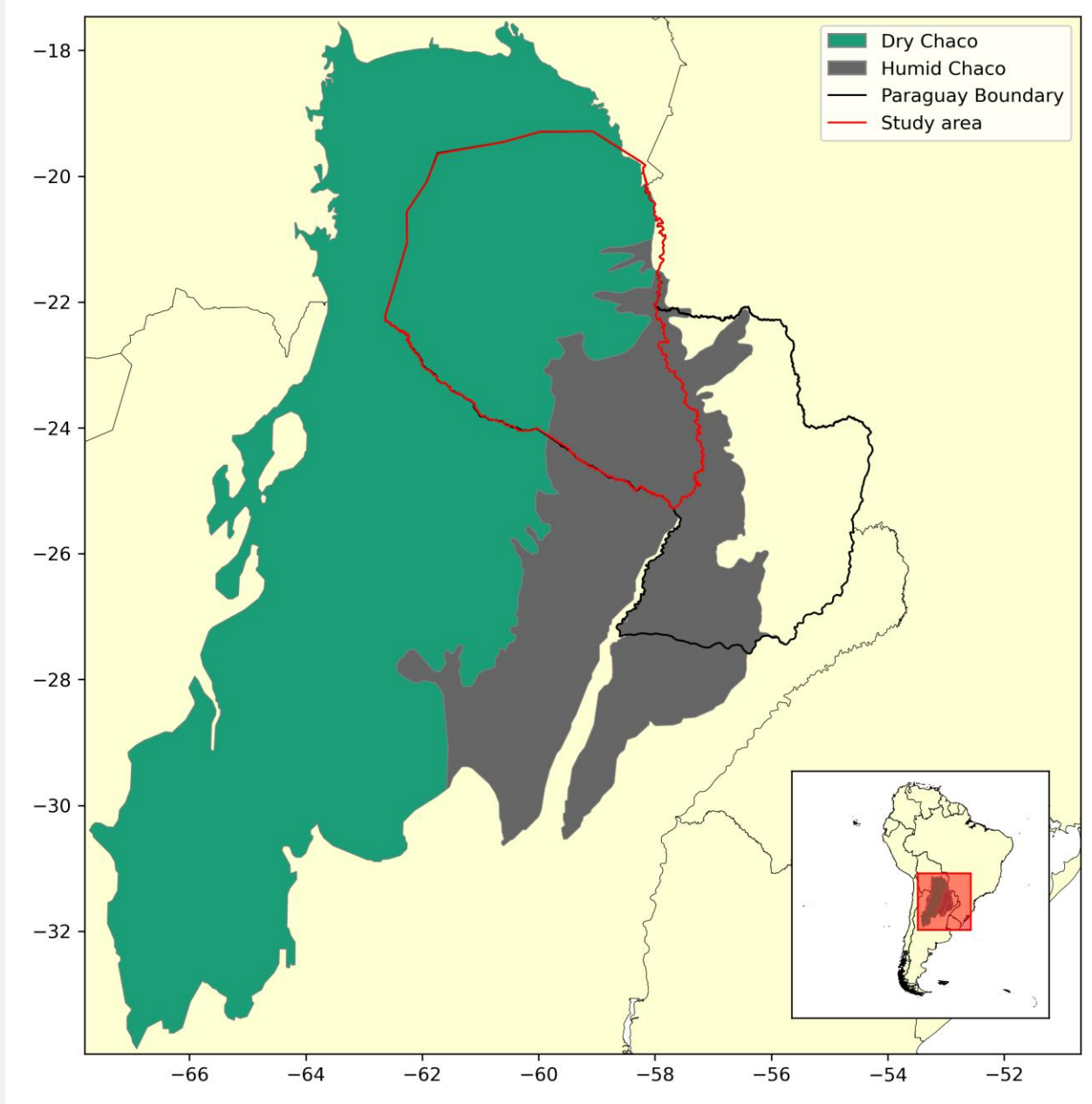
The main objectives of the project are:

- Develop a Random Forest model to predict daily wildfire likelihood in the Paraguayan Chaco
- Identify the most determinant explanatory variables of wildfire events in the Paraguayan Chaco region
- Provide proper validation metrics

STUDY AREA

This project will be focused on the Paraguayan Chaco, encompassing the dry and humid Chaco subregions.

The South American Gran Chaco ecoregion is the largest continuous tropical dry forest of South America, extending over Argentina, Brazil, Bolivia, and Paraguay.



DATA

Observation dataset: Fire Incidents

FIRMS MODIS (2015-2024): Daily fire detections from Moderate Resolution Imaging Spectroradiometer providing georeferenced wildfire occurrence data across the Paraguayan Chaco region.

Environmental Predictors

Topography - SRTM

Digital Elevation Model: Elevation, slope, and aspect

Climate Variables - Daily temporal resolution:

CHIRPS: Precipitation

NOAA CFSV2: Maximum/minimum temperature, potential evaporation, wind speed (u/v components) (6-hour intervals, aggregated to daily)

Vegetation & Land Surface:

MODIS MOD11A1: Land surface temperature (1km, daily)

Landsat: NDVI (30m, 32-day composite)

MODIS MCD12Q1: Land cover classification (500m, annual)

Proximity Variables:

Distance to roads (Paraguayan National Statistics Institute)

Distance to water bodies (Paraguayan National Statistics Institute)

Degree of urbanization (JRC GHSL)

All spatial data accessed and preprocessed through Google Earth Engine, resampled to 500m resolution for model consistency.

METHODS: DATA PREPARATION

Fire Event Data

MODIS FIRMS fire detection data (2015-2024) were obtained and spatially clipped to the Paraguayan Chaco region, resulting in 544,034 fire observations. Each fire event was assigned a unique identifier and retained essential attributes: date, coordinates, and fire status ("yes").

Fire Absence Generation

Non-fire observations were generated using stratified random sampling. For each year in the study period (2015-2024), 4,000 spatially random points were created within the Chaco boundary using **GeoPandas'** sample_points() method. Random dates were assigned to each point using a custom function accounting for month length and leap years, ensuring temporal distribution matched the fire event dataset. This process generated 40,000 non-fire observations, resulting in a final dataset of 584,034 observations.

Explanatory variables extraction

Predictor variables were extracted at each observation point location using Python's Earth Engine API and batch processing workflows to manage computational constraints.

Climate Variables (6-day temporal aggregation):

- Precipitation (CHIRPS Daily): Mean of preceding 6 days
- Wind speed (NOAA CFSV2): Calculated from U and V components using $\sqrt{U^2 + V^2}$, averaged over 6 days

METHODS: DATA PREPARATION

- Maximum and minimum air temperature (NOAA CFSV2): 6-day averages
- Land surface temperature (MODIS MOD11A1): Most recent 8-day composite
- Potential evaporation (NOAA CFSV2): Most recent 8-day value
- *Vegetation and Land Cover*
- NDVI (Landsat 32-day composite): Most recent monthly composite
- Land use classification (MODIS MCD12Q1): Annual UMD classification
- Degree of urbanization (JRC GHSL): Settlement density classification

Proximity Variables:

Distance rasters were generated in ArcGIS Pro using Euclidean distance calculations from:

- Road networks (Paraguayan National Statistics Institute)
- Water bodies (Paraguayan National Statistics Institute)

MODEL TRAINING

The consolidated dataset was split into training (80%, n=467,227) and validation (20%, n=116,807) sets using stratified sampling to maintain class distribution. The binary target variable was encoded as 1 (fire) and 0 (no fire).

The model was trained using Scikit-learn with the following optimized hyperparameters:

- Number of trees: 100
- Maximum tree depth: 20
- Minimum samples per split: 10
- Class weighting: Balanced
- Random state: 42

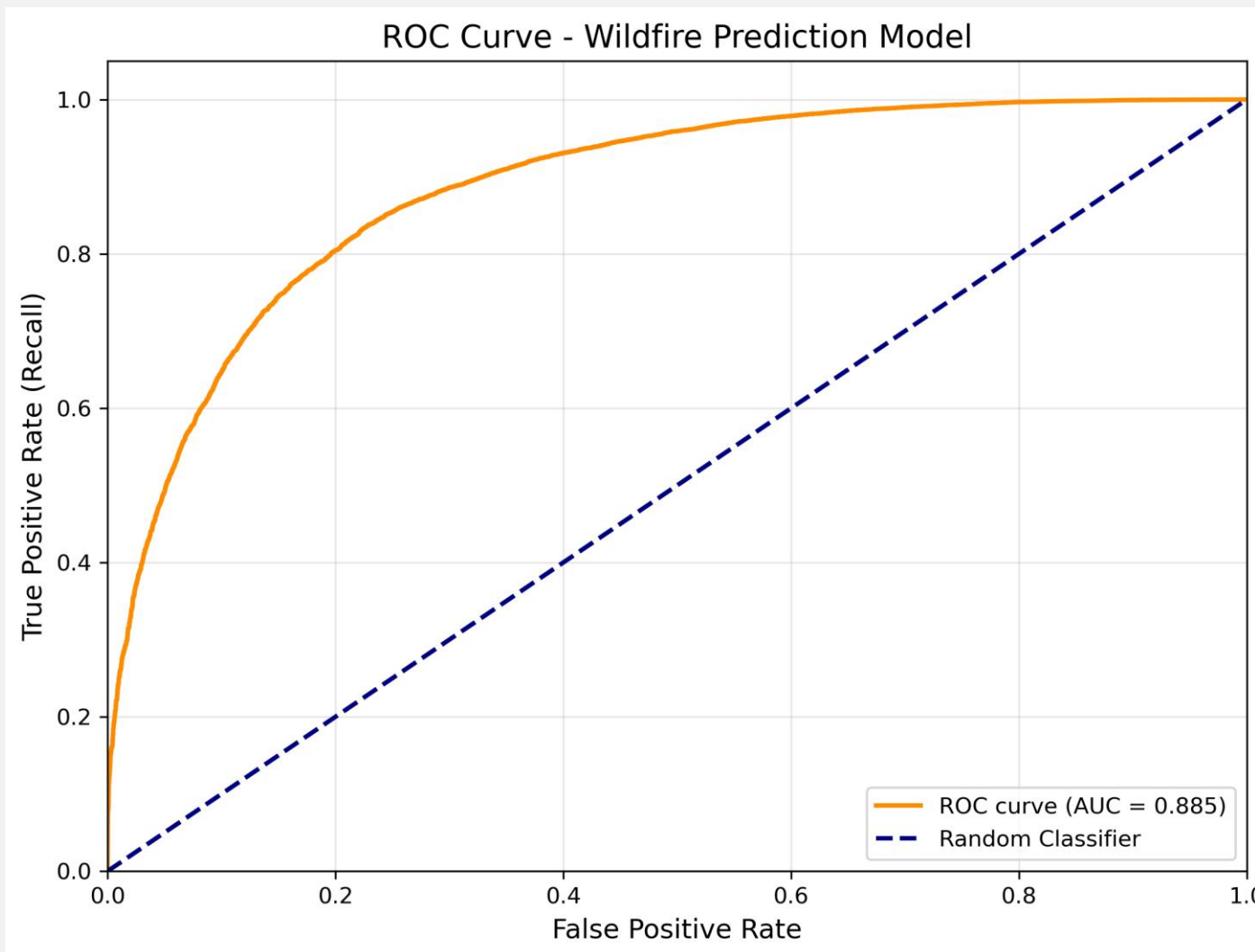
Hyperparameters were initially explored through 3-fold grid search cross-validation across 96 parameter combinations, with manual refinement to prioritize balanced performance between fire and non-fire detection.

MODEL EVALUATION

The Random Forest model demonstrated strong predictive performance on the validation dataset:

- Accuracy: 89.1%
- ROC-AUC: 0.885

The model captures 95.3% of actual fire events

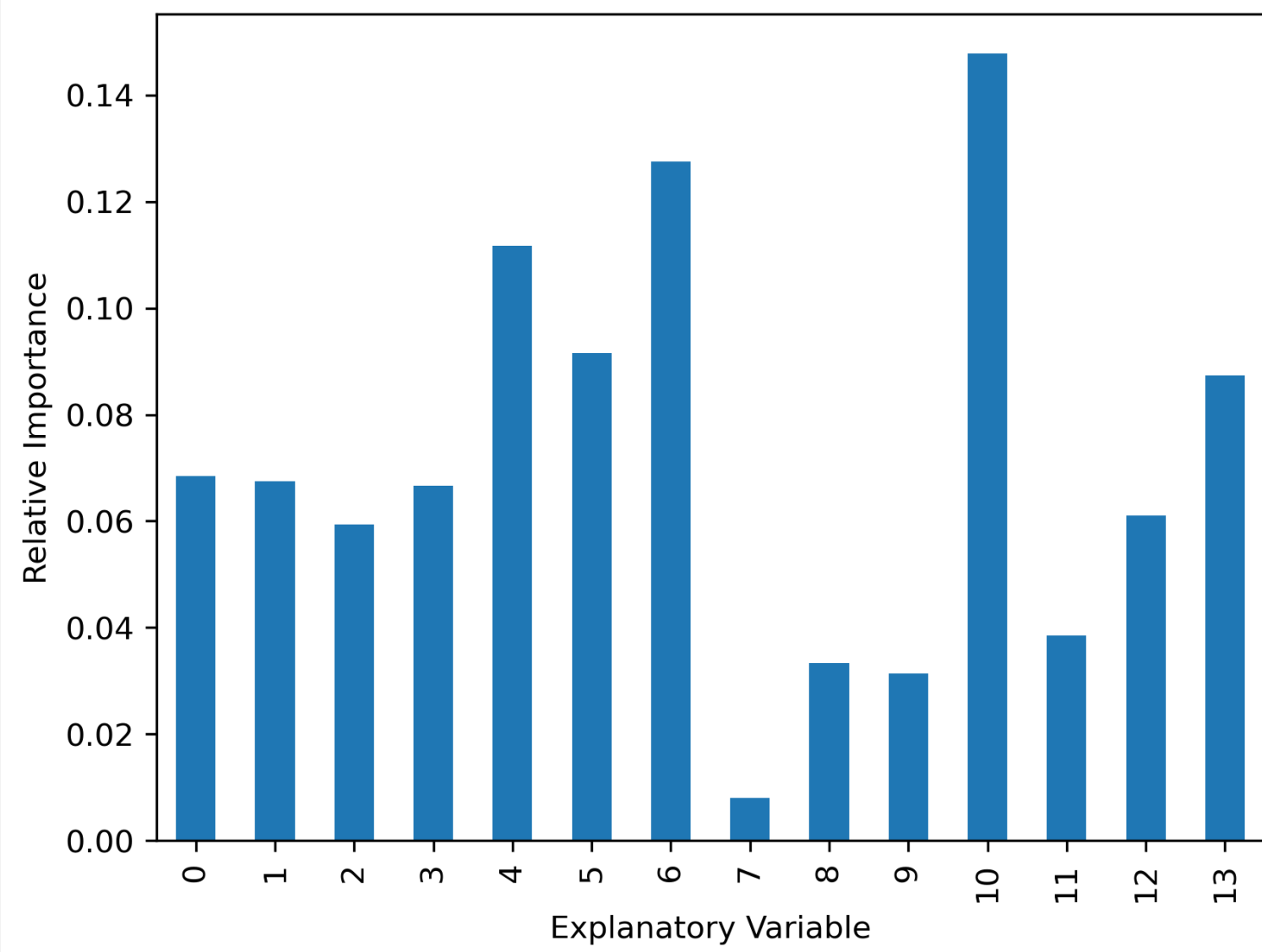
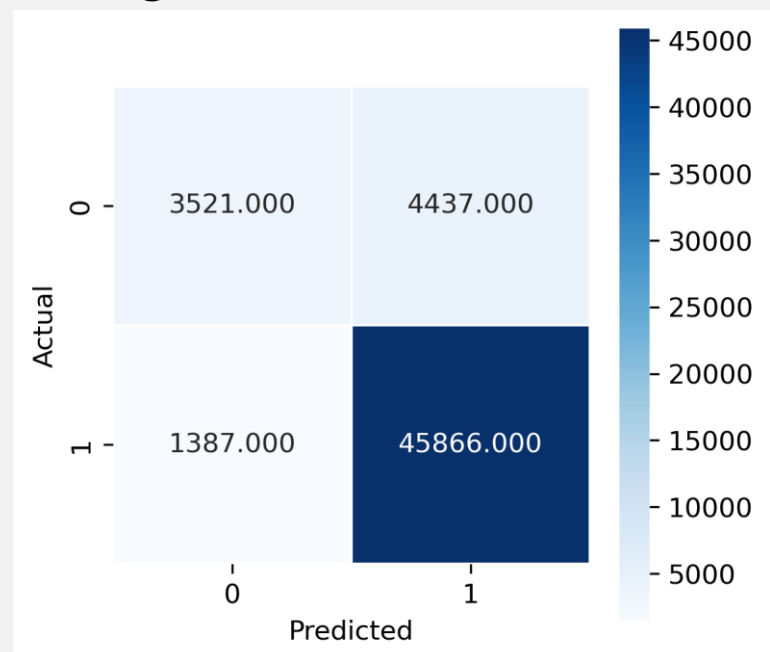


MODEL EVALUATION

Temperature variables and evapotranspiration collectively account for ~45% of predictive power.

Land use classification (1%) and degree of urbanization (<1%) showed negligible importance, possibly due to relatively homogeneous land cover in the study area and the low population density of the region.

Only 4.7% of fires are missed 1,387 out of 47,253).



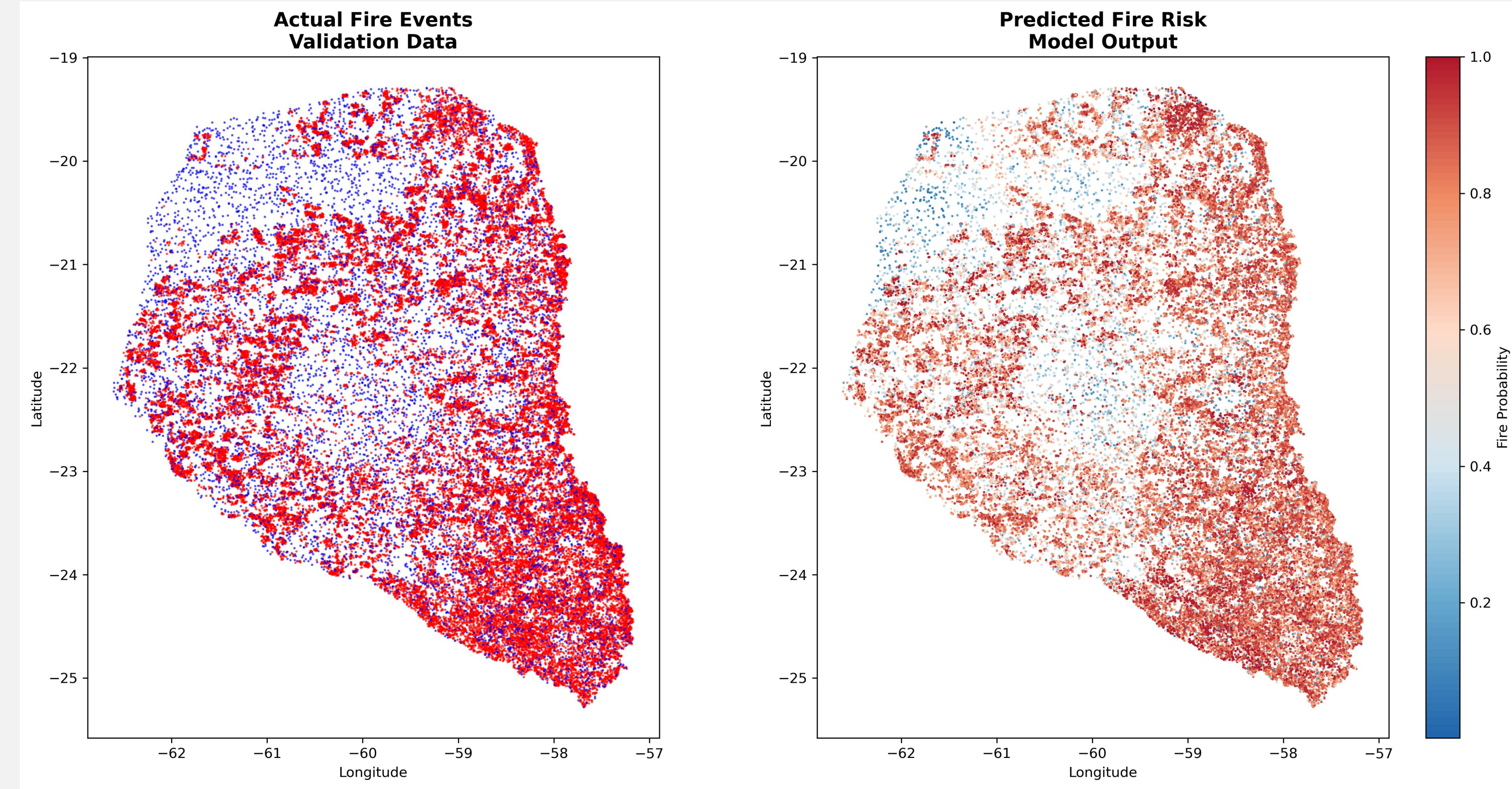
RESULTS AND DISCUSSION

The model demonstrated a slight bias toward fire detection, with 2,234 missed fires (4.7% false negative rate) versus 3,786 false alarms (47.6% false positive rate among non-fire events).

Key predictors of fire risk

Climate-driven patterns: Temperature extremes, potential evaporation, and land surface temperature collectively accounted for ~45% of predictive power. *Vegetation stress:* NDVI ranked third in importance, suggesting that drought-stressed vegetation significantly increases fire risk. *Anthropogenic influence:* Proximity to roads and water bodies (16% combined importance) indicates human activity and accessibility influence fire occurrence patterns, likely through both ignition sources and detection/suppression access.

Model output:



CONCLUSION

The random forest model effectively predicts fire events. It has a slight bias towards fire event prediction, which can limit the reliability of the model. To improve the current model, the following can be done:

- Increase the number of absence fire points – to a ratio of (2:1)
- Drop non-significant variables, particularly
- Map uncertainty given the stochastic nature of fire events

Overall, the model succeeds to predict fire events based on the climatic variables associated to fire events.